

L. Tharindu Nirmal Wickremasinghe

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Research: Generative AI, World models, Physically consistent video/image generation and restoration, Computational imaging.

EDUCATION

- PhD, Electrical & Computer Engineering** **GPA: 4.00/4.00** *May 2029*
Purdue University, USA · [Intelligent Imaging Lab](#), advisor: Prof. Stanley H. Chan
- MS, Electrical & Computer Engineering** **GPA: 4.00/4.00** *Dec 2026*
Purdue University, USA · Intelligent Imaging Lab
Inductee, Phi Kappa Phi National Honor Society – Purdue chapter (Top 10% graduate student).
- B.Sc.(Eng.), Electronics & Telecommunication Engineering** **GPA: 4.16/4.20** *Dec 2023*
University of Moratuwa, Sri Lanka · First Class Honours · Dean's List, all semesters
Ranked **1/118** (Gold Medal) in the graduating Electronics & Telecommunication class.
Ranked **1/980** (Gold Medal) across the Faculty of Engineering.

GRADUATE RESEARCH & PUBLICATIONS

- FlowSteer: Conditioning Flow Fields for Consistent Image Restoration** [CVPR'26 project page](#) 
Physics-consistent, zero-shot restoration with flow-matching models *Apr – Nov 2025*
 - Pioneered a **zero-shot** restoration framework that steers pretrained **flow models** across multiple degradation tasks with **no task-specific retraining**, resolving the usual fidelity–realism tradeoff: high **pixel-level fidelity** alongside high **perceptual quality**.
 - **Tharindu Wickremasinghe**, Chenyang Qi, Harshana Weligampola, Zhengzhong Tu, Stanley H. Chan. *FlowSteer: Conditioning Flow Field for Consistent Image Restoration*. CVPR, 2026.
- SeeU: Seeing the Unseen World via 4D Dynamics-Aware Generation** [CVPR'26 project page](#) 
Physically consistent video restoration through a 4D world model *Aug – Dec 2025*
 - Built a **2D → 4D → 2D** pipeline that lifts 2D inputs into a 4D scene, models its dynamics, and reprojects to video – a **new information flow and training scheme** that anchors generation in a realistic spatial and temporal context for **coherent** results.
 - Yu Yuan, **Tharindu Wickremasinghe**, Zeeshan Nadir, Xijun Wang, Yiheng Chi, Stanley H. Chan. *SeeU: Seeing the Unseen World via 4D Dynamics-aware Generation*. CVPR, 2026.
- NewtonGen: Text-to-Video Generation via Neural Newtonian Dynamics** [ICLR'26 project page](#) 
Physics-consistent and controllable text-to-video generation *May – Oct 2025*
 - Developed a controllable text-to-video framework embedding **learnable Newtonian dynamics** directly in generation, and devised **novel metrics** that track subject dynamics across distinct motion types – giving interpretable **physics knobs** over generated motion.
 - Yu Yuan, Xijun Wang, **Tharindu Wickremasinghe**, Zeeshan Nadir, Bole Ma, Stanley H. Chan. *NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics*. ICLR, 2026.

SELECTED PRIOR RESEARCH

- Affine Subspace Models and Clustering for Patch-Based Image Denoising** [IEEE ACSSC'25](#) 
IEEE ME-UYR young researcher program – mentor: Prof. Marco F. Duarte *Jan – Fall 2024*
 - Introduced **affine subspace clustering** for image-patch subspace models, improving the patch-based denoising baseline by **15%**; presented at **ICASSP 2025**, full paper accepted at the **Asilomar Conference on Signals, Systems and Computers 2025**.
- Masked Optimization with Selective Attention for Image Reconstruction (MOSAIC)** [Preprint](#) 
Undergraduate research with the [Wadduwage Lab](#), Harvard University *Nov 2022 – 2024*
 - Established the **critical role of attention** in adaptive image reconstruction for computational microscopy.
 - First-author extended abstract and conference presentation, *Lost in Translation: The Critical Role of Attention in Adaptive Image Reconstruction*, **OSA Optica Imaging Congress**, Cambridge, MA, 2023.

Real-Time Configuration of Intelligent Reflecting Surfaces

 [Video](#)  [Report](#)

IEEE Signal Processing Cup 2021 – supervised by Dr. P. Dharmawansa

Feb – Jun 2021

- Won **1st place worldwide** for a real-time IRS configuration solution and was invited to present at **IEEE ICASSP 2021**, Toronto; the competition and our solution were featured in *IEEE Signal Processing Magazine* 39(1), Jan 2022, pp. 126–131.

HPGNN: Hierarchical Graph Neural Networks for Point Cloud Processing

May – Dec 2021

Co-developed a **hierarchical GNN** aggregating multi-scale geometric context for outdoor point clouds; **26th Int. Conf. on Pattern Recognition (ICPR) 2022**, Montreal, pp. 2700–2706.

Real-Time Traffic Sign and Traffic Light Detection on Embedded Systems

Fall 2020 – 2021

Co-developed an embedded-deployable detector **enabling on-vehicle real-time inference**; **IEEE Intelligent Vehicles Symposium (IV) 2022**, Aachen, pp. 723–728.

AWARDS, SCHOLARSHIPS & INVITED PROGRAMMES

- **Gold Medal – Most Outstanding Graduand**, Faculty of Engineering, University of Moratuwa. *2023*
- **UNESCO Gold Medal** – best academic performance **across all faculties**, University of Moratuwa. *Dec 2023*
- **9 medals and awards** won at graduation- Faculty of Engineering, University of Moratuwa. *2018 – 2023*
- **Pre-doctoral invitee**, Cornell–Maryland–Max Planck Research School (CMMRS), Germany. *Aug 2023*
- **SGI 2023** – selected to audit the Summer Geometry Initiative workshops at MIT, MA, USA. *Aug 2023*
- **1st place**, IEEE PES design competition; Sri Lanka's IEEE CAS Region 10 representative – *Smart Breadboard. 2020/21*
- **Most Outstanding Student**, graduating class of 2017, Nalanda College, Colombo, Sri Lanka. *2017*
- **7 International Olympiad medals**, in Mathematics, Science, and Earth Science Olympiads. *2009 – 2015*
- **5 National Olympiad medals**, in Physics, Astronomy, Astrophysics, and Chemistry. *2012 – 2015*
- **National rank 1** (1/400,000), G.C.E. Ordinary Level examination, Sri Lanka. *2014*

RESEARCH & PROFESSIONAL EXPERIENCE

- **Purdue University, USA** *Aug 2024 – present*
Graduate Research Assistant; Teaching Assistant for ECE 301: Signals and Systems (Fall 2024).
Peer reviewer for CVPR, ECCV, ICML, NeurIPS, ICPR, ICASSP, ISCAS, and ACM ISS.
- **University of Moratuwa, Sri Lanka** *2022 – 2024*
Junior Instructor for Machine Vision (Dec 2023 – Aug 2024).
TA for Digital Signal Processing, Electronic Control Systems, and Laboratory Practice (2022 – 2023).
- **Harvard University-Center for Advanced Imaging** – Remote Undergraduate Researcher *2022 – 2023*
- **University of Sydney-School of Computing, Australia** – Research Intern Scholarship *2021 – 2022*

TECHNICAL SKILLS

Languages	Python (NumPy, Pandas, scikit-learn, PyTorch, TensorFlow), MATLAB, C++, HTML, L ^A T _E X
Generative AI	Diffusers, Transformers, Flux, Hugging Face, LoRA Fine-tuning, SFT/Post-training, CUDA, OpenCV
Infrastructure	Slurm, multi-GPU training, Linux, Git, Bash
Hardware	Solidworks, Altium, Multisim, LTSpice, Webots

HACKATHONS, IDEATHONS, AND DATASCIENCE COMPETITIONS

- Champions** [Datathon 2022](#) – telecom next-best-action recommender · [Data Storm 2021](#) – customer status prediction and product recommendation · [Belt & Road Electricity Simulation 2020](#) – energy-efficient power distribution · [Decrypt 2019](#) – driver health-emergency alerting app
- Runners-up** [Datathon 2023](#) – anomaly detection of suspicious website activity

SERVICE, OUTREACH & INTERESTS

- **Teaching** – Lab outreach- machine learning for high school students (Summer 2025).
Instructor at “Algebra by 7th” Minority Education program at Purdue (Spring 2025).
- **Service** – Co-founder, Danuma Yathra Foundation. Taught high-school mathematics and science in rural Sri Lankan schools facing severe STEM teacher shortages 2018 – 2022.
 - Assistant Manager, Knowledge Hub of ENTC 2021 – 2022.
 - Rotaractor – University of Moratuwa 2018 – 2023.

- **Public Communication** – Panel moderator at “*Future of AI in Engineering*” at EXMO 2023, University of Moratuwa’s flagship public engineering exhibition.
- Poster at *Communicating Astronomy with the Public* (CAP) 2022 on organizing and volunteering amateur astronomers.
- Volunteer instructor, NCAS Astronomical Society 2018 – 2022.
- **Hobbies** – Chess, Badminton, Hiking, Photography.